Lab 1 Report: Introduction to Basic Electrical Tools

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Part One: Lab Safety Rules and Regulations (3 pts)

Read and understand the Laboratory Rules and Regulations documents available on Blackboard.
By signing below, I affirm that:

[1] I have understood the Laboratory Rules and Regulations.

[2] I agree to comply fully with all Laboratory Rules and Regulations.

[3] I understand that any violation of the Laboratory Rules and Regulations may affect my final course grade.

Signature: 谢辰晨 熊润野

Date: 2026.9.24

Part Two: Internal connections of the Breadboard

Breadboard is used to make up temporary circuits for testing or to try out an idea. No soldering is required, so it is easy to change connections and replace components. Breadboards come in various brands and sizes to suit the requirements of electronic circuits. Nevertheless, all breadboards share the same fundamentals.

1. Read the document about using breadboards.

2. Connections on breadboard

Breadboards have many tiny sockets (called 'holes') in them and strips of metal below the board which connect the holes placed on the top of the board. All "holes" that are electrically connected are said to share a common node.

As shown in Figure 1, the top two rows (A or D) of the board are internally connected sideways, and the holes in groups B and C are connected vertically.

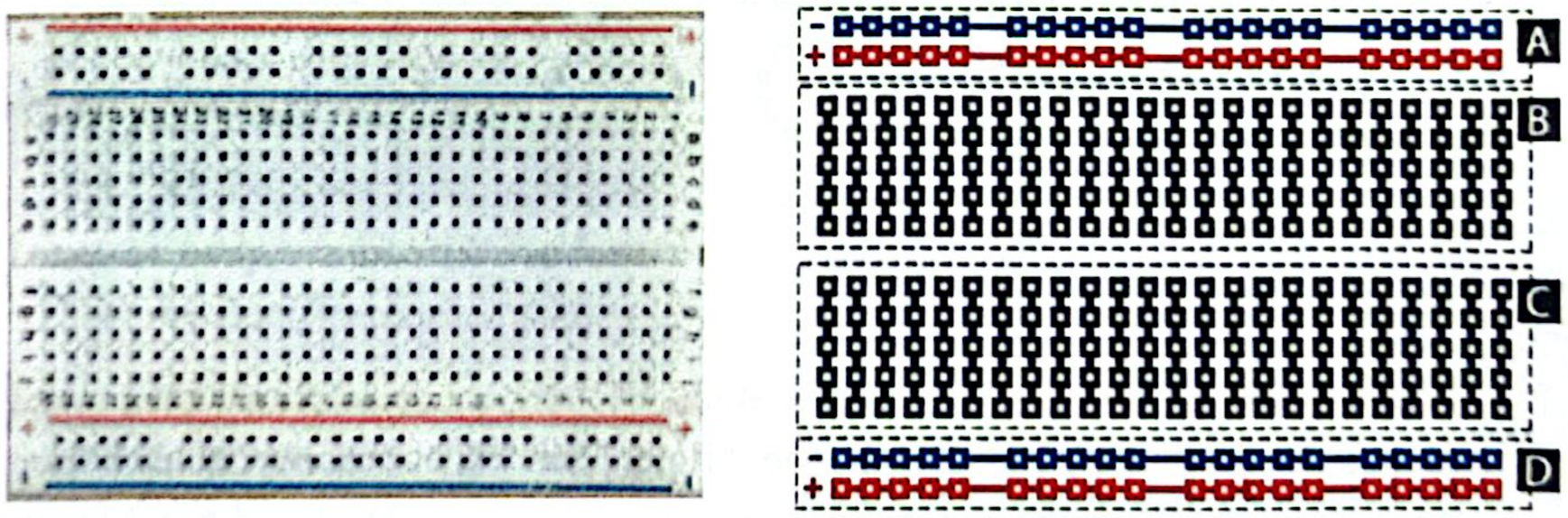


Figure 1

Use the breadboard shown in Figure 2 as a model for your breadboard. Draw lines connecting "holes" which are electrically connected. (6pts)

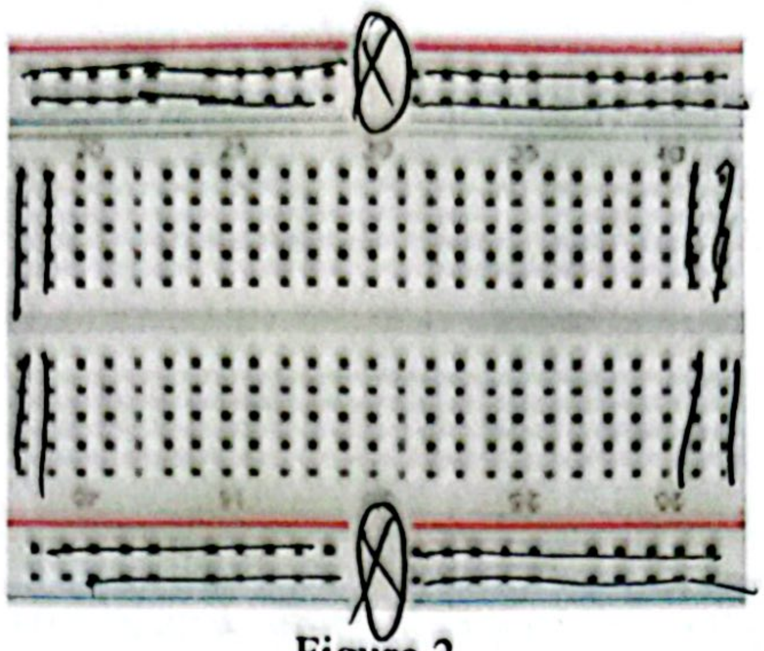


Figure 2

3. Internal connections of the Breadboard TEST Using DMM

CONTINUITY TEST is used to check if two points are electrically connected. If continuity exists, the beeper sounds continuously.

Insert the black test lead into the COM terminal, and the red test lead in the terminal labeled  . Put the DMM in continuity mode  by pressing the button  .

1) Please connect the wires as shown in Figure 3.

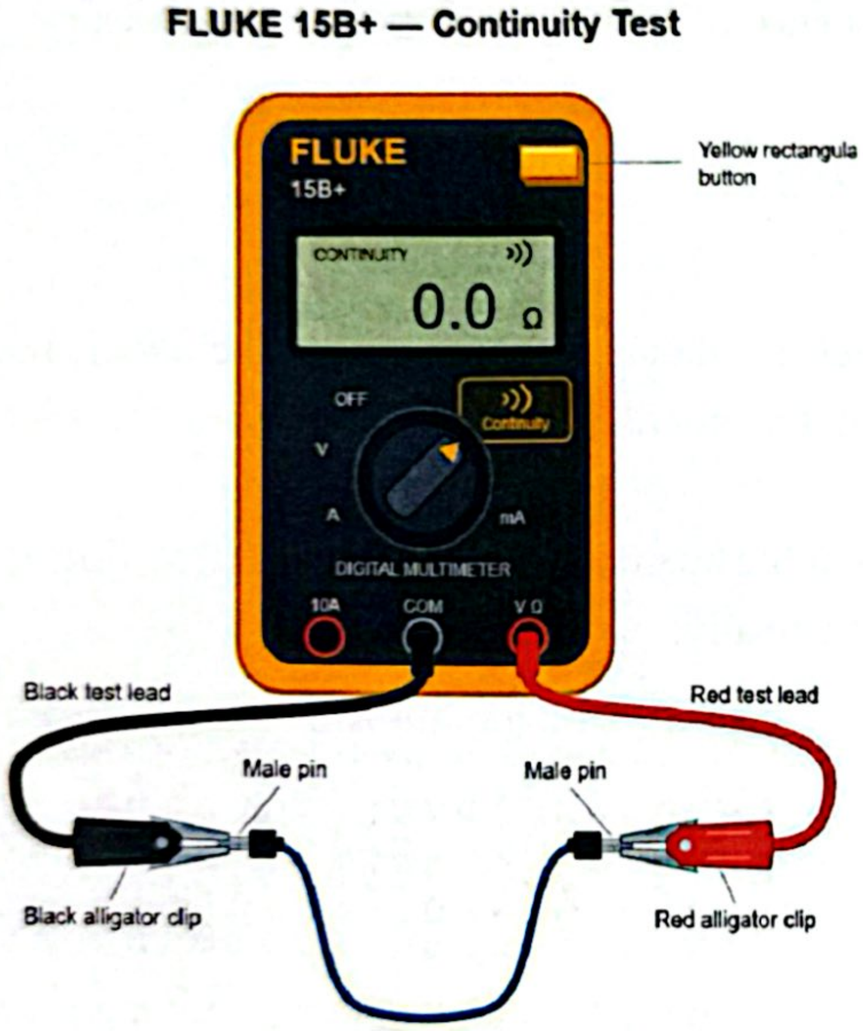


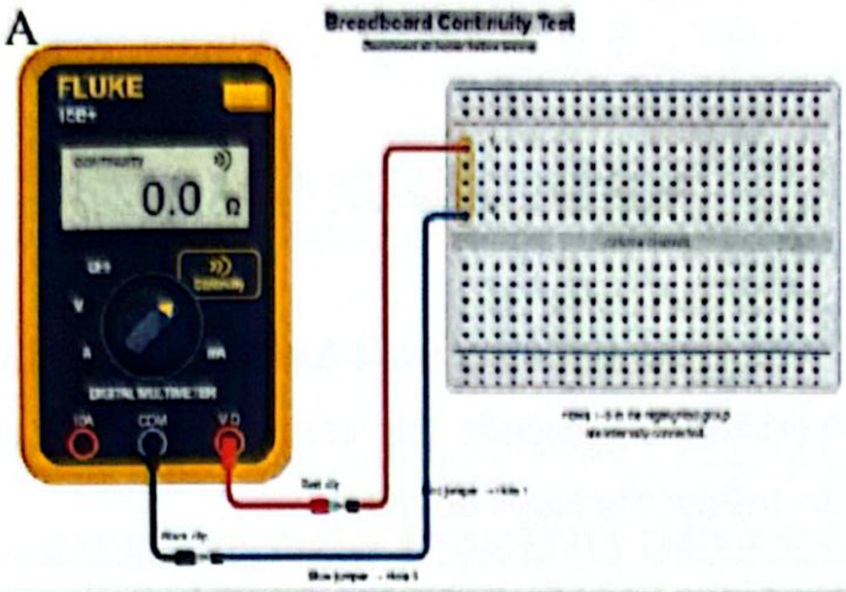
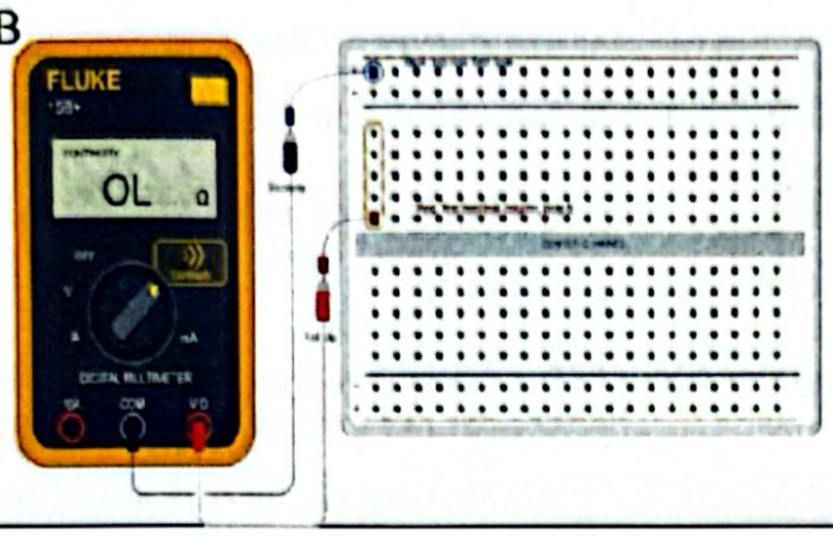
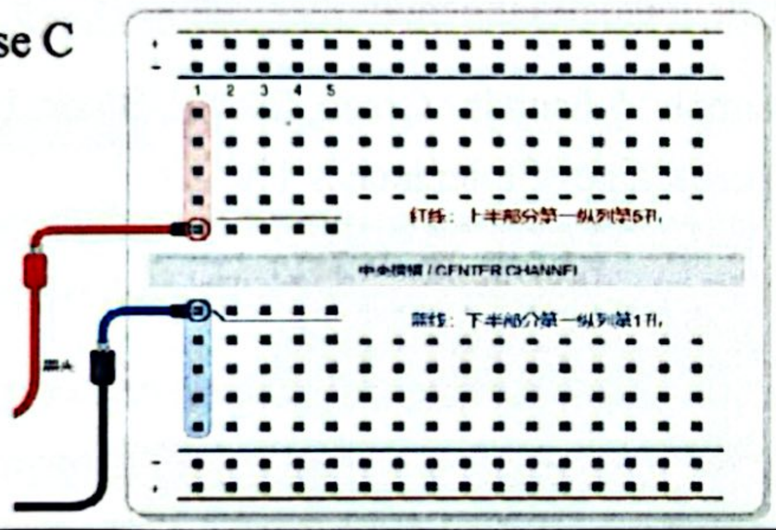
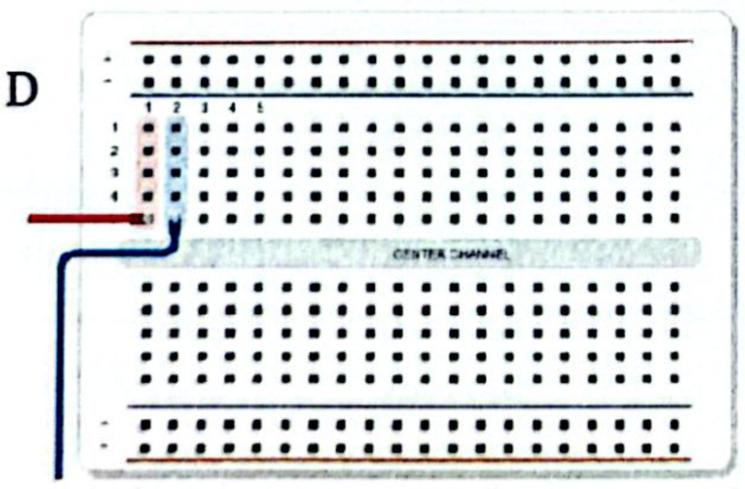
Fig.3

What did you hear? Beep.

Ask for help if you heard nothing. (2pt)

2) A breadboard is provided at your lab station. "Holes" that are electrically connected are **short** and said to share a common node. "Holes" that are electrically isolated from each other are **open**. Using the continuity testing function of the DMM and two jumper wires can determine which "holes" on the breadboard are connected and which are electrically isolated from each other. Connect a jumper wire to each of the alligator clips. Then put a jumper wire in one hole of a breadboard, and put another jumper wire in another hole. Complete the table. (8 pts, 1 pt each)

Table 1

	Continuity Measurement Beeping?(Yes or NOT)	resistance between "holes"
Case A 	Yes	1.0 Ω
Case B 	NOT	1
Case C 	NOT	1
Case D 	NOT	1

Part Three: Resistor Color Code and Measurement of Resistance

Color-coded resistor and Resistor Color Code

The resistor is the most basic and widely used component in electronic circuits. And the relation between the voltage and the current of a resistor in a circuit will follow Ohm's law.

As shown in Fig. 4, there are two types of color coding: 5-band-code and 4-band-code resistors.

The color bands on the body of the color-coded resistor tell how much resistance it has.

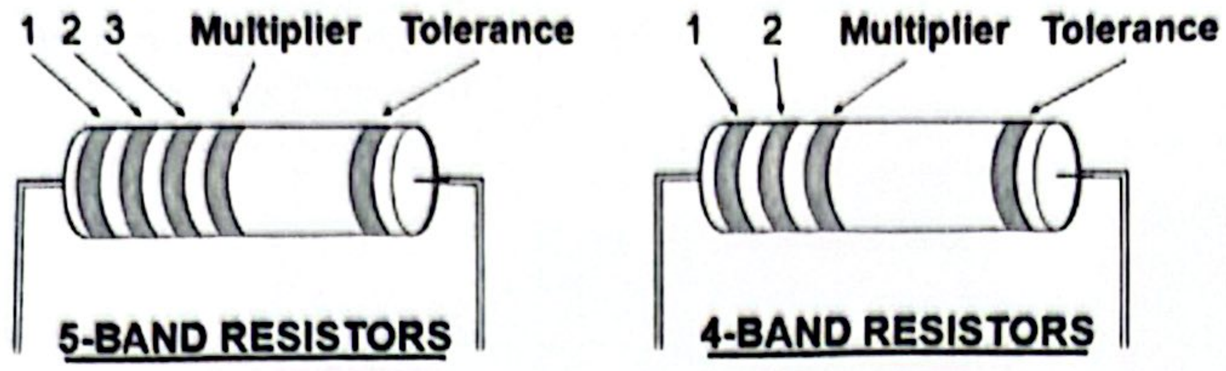


Fig.4

As can be seen, the 5-band code has one more digit resolution than the 4-band code. The following chart provides the color code for a 5-band resistor. To decode the color bands and calculate the correspondingresistance value, one needs to follow the steps below.

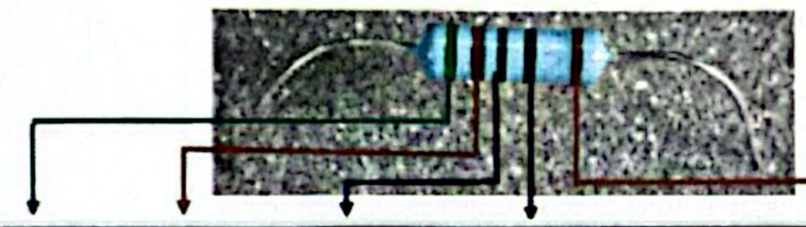
1) Find the tolerance band. As shown in Fig. 4, for both 5- and 4-band resistors, the last band indicates tolerance. It is located at one end of the resistor and far away from therest of the color bands. It gives the accuracy of the actual resistance to the value thatis labeled.

2) Start from the other end and use the color code map to identify the color band. Thiswill be the first digit (the most significant digit) of the resistance value.

3) Then, similarly, decode the second and the third band (for 5-band resistors only). Writedown all the digits in order (from left to right).

4) The next band is the Multiplier band, which indicates the multiplier factor.

Take the 5-band-code resistor shown in Fig. 5 for example. 5-band is : Green, Brown, Black, Black, Brown, so the resistance is 510*100=510Ω and the percentage of tolerance is 1%



COLOR	BAND 1	BAND 2	BAND 3	BAND 4 MULTIPLIER	BAND 5 TOLERANCE
BLACK	—	0	0	$\times10^0 = \times1$	—
BROWN	1	1	1	$\times10^1 = \times10$	±1%
RED	2	2	2	$\times10^2 = \times100$	±2%
ORANGE	3	3	3	$\times10^3 = \times1k$	—
YELLOW	4	4	4	$\times10^4 = \times10k$	—
GREEN	5	5	5	$\times10^5 = \times100k$	±0.5%
BLUE	6	6	6	$\times10^6 = \times1M$	±0.25%
VIOLET	7	7	7	$\times10^7 = \times10M$	±0.1%
GREY	8	8	8	$\times10^8 = \times100M$	±0.05%
WHITE	9	9	9	$\times10^9 = \times1G$	—
GOLD	—	—	—	$\times10^{-1} = \times0.1$	±5%
SILVER	—	—	—	$\times10^{-2} = \times0.01$	±10%

Figure 5. Color Code

Questions: For a 1kΩ resistor, what are the color band colors and associated band values? (5 pts, 1 pt each)

Table 2

Resistor Value	Band1	Band2	Band3	Band4	Band5
1kΩ, 1%	Brown	Black	Black	Brown	Brown

What is the expected nominal resistance and tolerance (in Ohms)? (4 pts. 1 pt each)

R = 1000 Ω ± 10 Ω (not %)

R_{min} = 990 R_{max} = 1010.

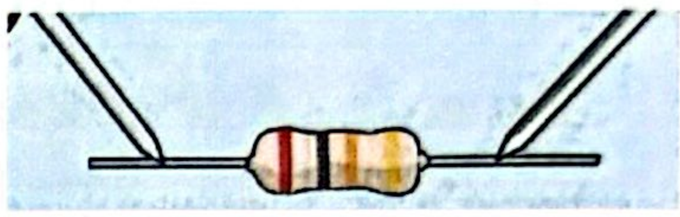
How to use a multimeter to measure resistance.

1) Firstly, turn the rotary switch to  . Make sure power is disconnected from the circuit to be measured.

2) Insert the black test lead into the COM  terminal, and the red test lead into the terminal labeled Ω  , which is the symbol for ohms.

3) Touch the tips of the probes onto each end of the resistor.

4) Read the measured resistance on the display.



Procedure

Select **three** 1kΩ resistors, and measure the resistance of each using the digital multimeter and compare the values with the specified value. (9 pts, 1 pt each)

The **percentage difference (% Error)** between the nominal value (indicated by the color code) and the measured value is calculated using the equation:

$$\textbf{percentage difference} = \left|\frac{Nomimal\ Value - Measured\ Value}{Nominal\ Value}\right| \times 100\%$$

Table 3

Resistor	Measured Value	percentage difference	Within Tolerance?
①-1kΩ	992 Ω	0.8%	Yes
②-1kΩ	993 Ω.	0.7%	Yes
③-1kΩ	994 Ω.	0.6%.	Yes

Part Four: Generate DC Voltage and Measurement of DC Voltage

DC Power Supply

Questions:

1. Read the document about using a DC power supply. What does a DC power supply do? (3pts)

Supply Direct Current electric power.

2. The DC power will be utilized for this laboratory experiment.

1) Please record the model number DH1766-1. (1pt)

2) Complete the table below. (6 pts, 2 pts each)

Table 4

Output Voltage Range (reference to GND)	Value
CH1	0~32V
CH2	0~32V
CH3	0~6V

Measurement of DC Voltage

1. Turn the DC power supply ON.
2. Set the DC power supply output voltage to 1.5 Volts, then switch the DC supply channel ON.
3. Use the DMM to measure the output voltage of the DC power supply.

Question: Sketch the power supply and the DMM FLUKE 15B+, including the position the **rotary switch** should be in, the connection of the red test lead, and the connection of the black test lead. (4 pts) Warning: while measuring voltage, the multimeter dial SHOULD NOT be in the mA (current) position. It might destroy/damage the mete.





4. Verify this voltage using a voltmeter.

In electronics and electrical engineering, we usually define a point in a circuit as a reference point. This reference point is known as ground (or GND) and carries a voltage of 0V. Voltage measurements are relative measurements. That is, a voltage measurement must be compared to another point in the circuit. If it is not, the measurement is meaningless. All voltage measurements are referenced with respect to ground (zero) unless otherwise noted. The

conventional approach is to use the black probe so that it is "closest" to the COM (which is the "ground" reference of the circuit), and to use the red probe so that it is "closest" to the voltage source (+1.5 volts in this circuit).

5. Record the voltage readings on the DMM (including sign) and the reading on the DC power supply's meter. (2 pts, 1 pt each)

Table 5

DC power supply	DMM
1.499V	1.498V

6. Record and explain the differences when the positive and negative leads are switched at the DMM. (4pts)

It's Breading becomes an opposite number. Because the input direction is changed.

Part Five: Measurement of a simple electric circuit

For the circuit shown in Figure 6, R1 = R2 = R3 = 1 kΩ.

In Figure 7, try to connect the resistors in series to the DC Power supply to make a complete circuit as shown in **Figure 6**. (6pts)

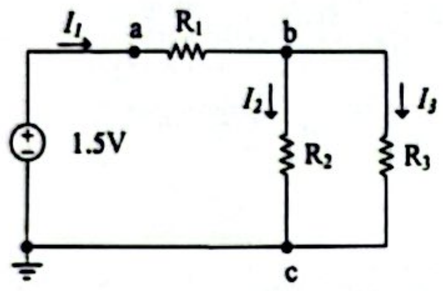


Figure 6

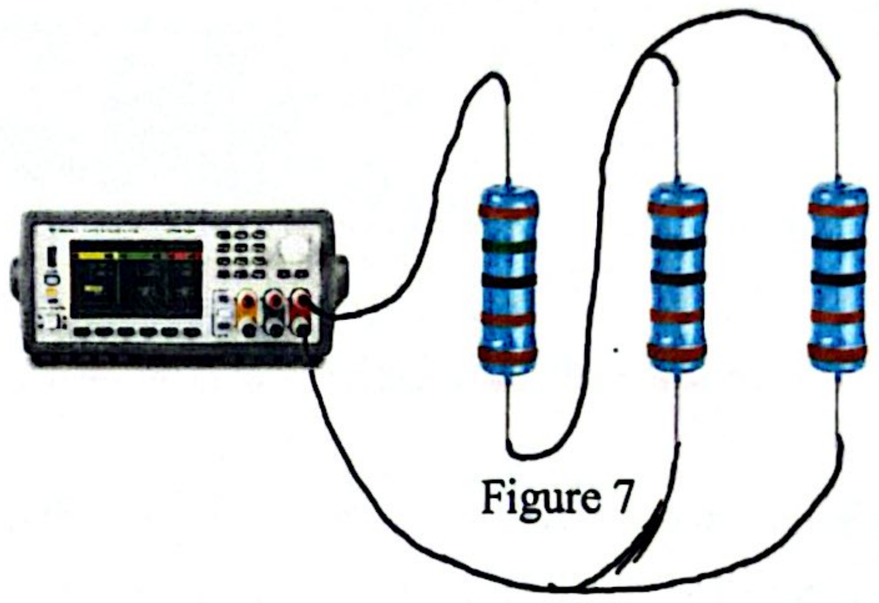


Figure 7

Measurement of DC voltage

1. In Figure 8, sketch the connection to measure the voltage Vab in Figure 6, including the position where the rotary switch should be. (6 pts)

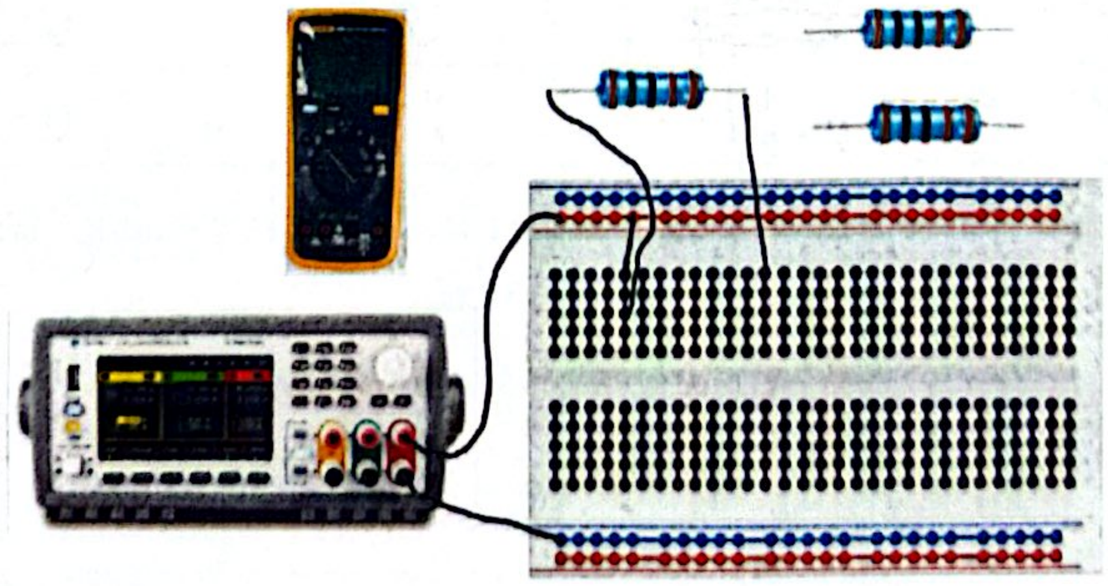


Figure 8

2. Use theoretical analysis to determine the expected voltage and record the values in Table 5 (4pts).

3. Build the circuit shown in Figure 6 on the breadboard. Measure the voltage with DMM, and record the values in Table6 (8 pts).

Table 6

	V_{ab}(V)	Vbc (V)	V_{ab}+ V_{bc}(V)	VS (V)
Theory	1	0.5	1.5	1.5
Measure	0.997.	0.499.	1.496.	1.498

Measurement of current

The DMM can also be used to measure current through a device. Measuring current is different from measuring voltage or resistance. To measure current, we must connect the DMM meter in ammeter mode in series along the branch in which we are measuring current.

1. Sketch the connection in Figure 9 below to measure the current I, including the position where the rotary switch should be. (6 pts)

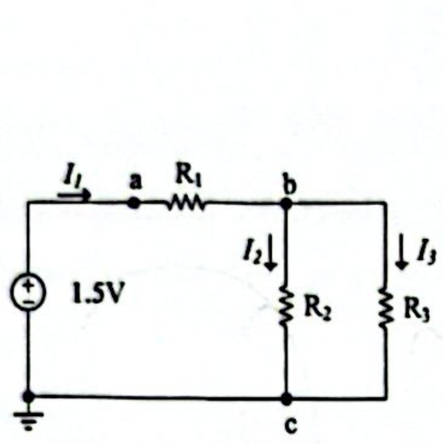


Figure6

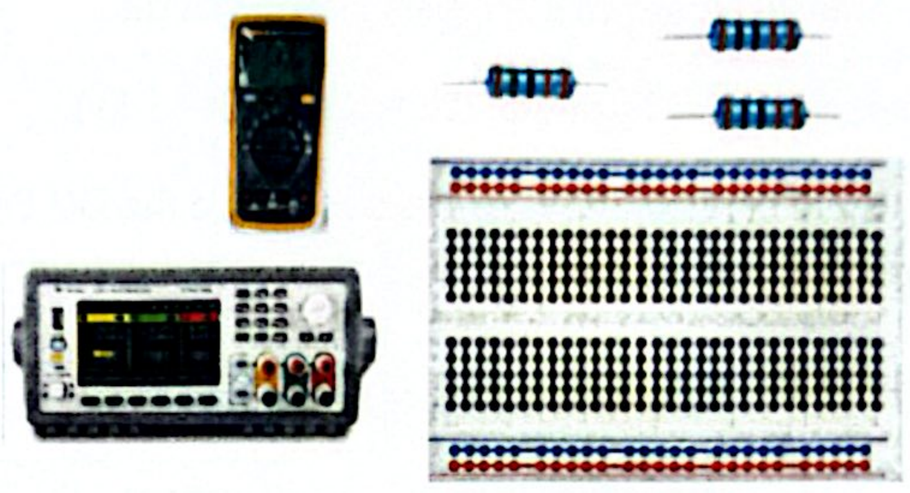


Figure 9

2. Use theoretical analysis to determine the expected current and record the values in Table 7 (3 pts).
3. Build the circuit shown in Figure 6 on the breadboard. Set up the DMM to measure DC current.
4. Complete the table below. (6 pts, 1 pt each)

Table 7

	I_1(mA)	I_2(mA)	I_3(mA)
Calculated values	1	0.5	0.5
Measured values	0.99	0.49	0.49

5. What is the sum of the currents I_2 and I_3? How does I_1 compare with I_2+ I_3? What circuit law is verified? (7 pts, the first two 2 pts, the last one 3 pts)

$I_2 + I_3 = I_1 = 1$ mA Kirchhoff's Current Law